

19.1

$$\frac{1}{30} + \frac{1}{s'} = -\frac{1}{15} \quad s' = -10\text{cm}$$

$$M = -\frac{-10}{30} = \frac{1}{3}$$

19.2

$$\frac{1}{f} = (n-1)\left(\frac{1}{10} - \left(-\frac{1}{10}\right)\right) \quad f = 9.6\text{cm}$$

$$\frac{1}{f} = (n-1)\left(-\frac{1}{10} - \frac{1}{10}\right) \quad f = -9.6\text{cm}$$

19.3

$$\frac{1}{f} = (0.2)\left(\frac{1}{\infty} - \left(-\frac{1}{57.1}\right)\right) \quad f = 285.5\text{mm}$$

$$\frac{1}{500} + \frac{1}{s'} = \frac{1}{286} \quad s' = 665.5\text{mm}$$

19.4

第一次成像：

$$\frac{1}{12} + \frac{1}{s'} = \frac{1}{8} \quad s' = 24\text{cm}$$

第二次成像：

$$\frac{1}{12} + \frac{1}{s''} = \frac{1}{6} \quad s'' = 12\text{cm}$$

$$M = \left(-\frac{24}{12}\right) \times \left(-\frac{12}{12}\right) = 2 \quad h = 2 \times 8 = 16\text{cm}$$

第二个透镜右侧 12cm；高 16cm

19.5

$$\frac{1}{8.5} + \frac{1}{s'} = \frac{1}{8.0} \quad s' = 136\text{mm}$$

$$\text{解法 1: } M = -\frac{136}{8.5} \times \frac{25}{4} = -100$$

$$\text{解法 2: } \mathbf{M} \approx -\frac{136}{8} \times \frac{25}{4} = -106$$

19.6

$$\frac{1}{s} + \frac{1}{21} = \frac{1}{1} \quad s = 1.05\text{cm}$$

解法 1: $\boldsymbol{M} = -\frac{21}{1.05} \times \frac{25}{2} = -250$ 倍

解法 2: $\boldsymbol{M} \approx -\frac{21}{1} \times \frac{25}{2} = -263$